

### 2.4.3 Artin–Schreier–Witt theory: order $p^r$

**Classes and equations.** For an  $\mathbb{F}_p$ -algebra  $A$ , write  $W_r(A) = A^r$  for the ring of  $p$ -typical Witt vectors of length  $r$ , with Witt addition and subtraction denoted by  $\oplus$  and  $\ominus$ , and put  $\wp = F - \text{id}$ . For a field  $K$  of characteristic  $p$ , write

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) := H_{\text{ét}}^1(\text{Spec } K, \mathbb{Z}/p^r\mathbb{Z}).$$

Its elements classify  $\mathbb{Z}/p^r\mathbb{Z}$ -torsors over  $K$ . Artin–Schreier–Witt theory gives a natural isomorphism

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) \cong W_r(K)/\wp W_r(K).$$

Under this isomorphism, the coset  $[\vec{g}] \in W_r(K)/\wp W_r(K)$  corresponds to an affine solution scheme as follows. Let  $\vec{X} = (X_0, \dots, X_{r-1})$  be the universal Witt vector on  $\mathbb{W}_{r,K} = \text{Spec } K[X_0, \dots, X_{r-1}]$ . Then

$$T_{\vec{g}} := \text{Spec}\left(K[X_0, \dots, X_{r-1}]/(\wp \vec{X} \ominus \vec{g})\right),$$

where the parenthesized Witt vector denotes the ideal generated by its  $r$  coordinate polynomials. Equivalently,  $T_{\vec{g}}$  is the fiber of  $\wp : \mathbb{W}_{r,K} \rightarrow \mathbb{W}_{r,K}$  over the  $K$ -point  $\vec{g}$ . Its first defining equation is  $X_0^p - X_0 - g_0 = 0$ , and each subsequent equation has the triangular form

$$X_j^p - X_j - g_j + C_j(X_0, \dots, X_{j-1}; g_0, \dots, g_{j-1}) = 0.$$

Here  $C_j$  is the universal Witt correction polynomial determined by the addition law and depends only on the preceding coordinates. This is a finite étale  $K$ -scheme of degree  $p^r$ , on which  $\vec{c} \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$  acts by Witt translation  $\vec{X} \mapsto \vec{X} \oplus \vec{c}$ . In particular, replacing  $\vec{g}$  by  $\vec{g} \oplus \wp \vec{h}$ , with  $\vec{h} \in W_r(K)$ , gives an isomorphic torsor. If  $[\vec{g}]$  has order  $p^s$ , its associated character cuts out a cyclic field extension of degree  $p^s$ ; thus  $T_{\vec{g}}$  is connected precisely when  $[\vec{g}]$  has order  $p^r$ . Conversely, every cyclic extension of degree dividing  $p^r$  arises in this way. The Witt-vector group scheme is constructed in [Section A.3](#). The Artin–Schreier–Witt operator, the proof that the displayed equation defines a  $\mathbb{Z}/p^r\mathbb{Z}$ -torsor, and the classification theorem—including its description of the degree, connected components, and the induced torsor under the subgroup  $\mathbb{Z}/p^s\mathbb{Z} \subseteq \mathbb{Z}/p^r\mathbb{Z}$ —are given in [Section A.4](#).

**Definition 52.** Let  $K = k((x))$ . For  $g \in K$ , put  $m(g) := \max(0, -v_x(g))$ . A vector  $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$  is **reduced** if, for every  $j$ , either  $m(g_j) = 0$  or  $p \nmid m(g_j)$ .

**Theorem 53** (Ramification in Artin–Schreier–Witt Extensions (Schmid–Witt)). *Let  $K = k((x))$ , with  $k$  perfect of characteristic  $p$ , and let  $L/K$  be a cyclic totally ramified extension of degree  $p^r$  whose Artin–Schreier–Witt class is represented by a reduced vector  $\vec{g} = (g_0, \dots, g_{r-1})$ , with  $m_j := m(g_j)$ . Then the largest upper-numbering ramification break of  $L/K$  is*

$$u = \max_{0 \leq j \leq r-1} p^{r-1-j} m_j.$$

The Schmid–Witt formula is an external ramification theorem: we use it as stated and do not reproduce its proof. For finite residue fields, the formula is due to Kanesaka–Sekiguchi [\[KS79\]](#) and Thomas [\[Tho05\]](#), building on Schmid [\[Sch37\]](#) for  $r = 1$ ; for arbitrary perfect residue fields, it is due to Elder–Keating [\[EK25\]](#). We use it in one direction only: evaluating the formula on one reduced representative computes the largest upper break, so no minimization over representatives is needed.

Reducedness prevents cancellation between Witt levels. Indeed, if two nonzero pole orders  $m_j$  and  $m_{j'}$ , with  $j < j'$ , contributed the same weighted value, then  $m_{j'} = p^{j'-j} m_j$  would be divisible by  $p$ , contradicting reducedness. For  $r = 1$ , the formula is exactly the Artin–Schreier break computation.